



REQUEST-TO-PAY SERVICE FINANCIAL INSTITUTION TECHNICAL SPECIFICATIONS

(API DESCRIPTION DOCUMENT)

Client	Banks, EMIs, PSPs
Version	2.1
Classification	Confidential

Version	Author(s)	Date	Modifications
1.0	IT	Sep-2019	Initial version
1.1	IT	Jan-2020	Addition of new JSON fields & Operations: 3.13,3.14,3.16, 3.17, 4.5, 5.5, 6.3, 6.4, 6.5
1.2	IT	Aug-2020	Update to Payer Service – List Operation
1.3	IT	Dec-2021	Updates to Process Flow, Biller Service and removal of Payer Service operations
1.4	IT	May-2025	Update to include request signature 3.14
1.5	IT	May-2025	Update of Request signature requirement 3.14 Update of Failing Scenarios of FTC request 5.2.3 .
1.6	IT	May-2025	Update to 5.1.5 Transaction Status possible values and Transaction Status reason codes possible values Update 5.2.5 Mandatory field reason code Update 5.2.7 update biller status
1.7	OPS	May-2025	Update 4.1.4 Query Update 4.1.5 Query Transaction Status Update 4.1.6 List Update 4.2.5 Notification from Payer's institution to GhIPSS
2.0	OPS	June-2025	Review and validation of Updates 1.6 and 1.7 and document released for publication.
2.1	OPS	June-2025	Re-arranged fields in the sample request payload alphabetically for the generation of request signature.

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1. Introduction

This document contains the technical specifications for the implementation of the REQUEST-TO-PAY SERVICE by Financial Institutions. It provides the definitions, communication protocols, and tools (API) for implementing the service.

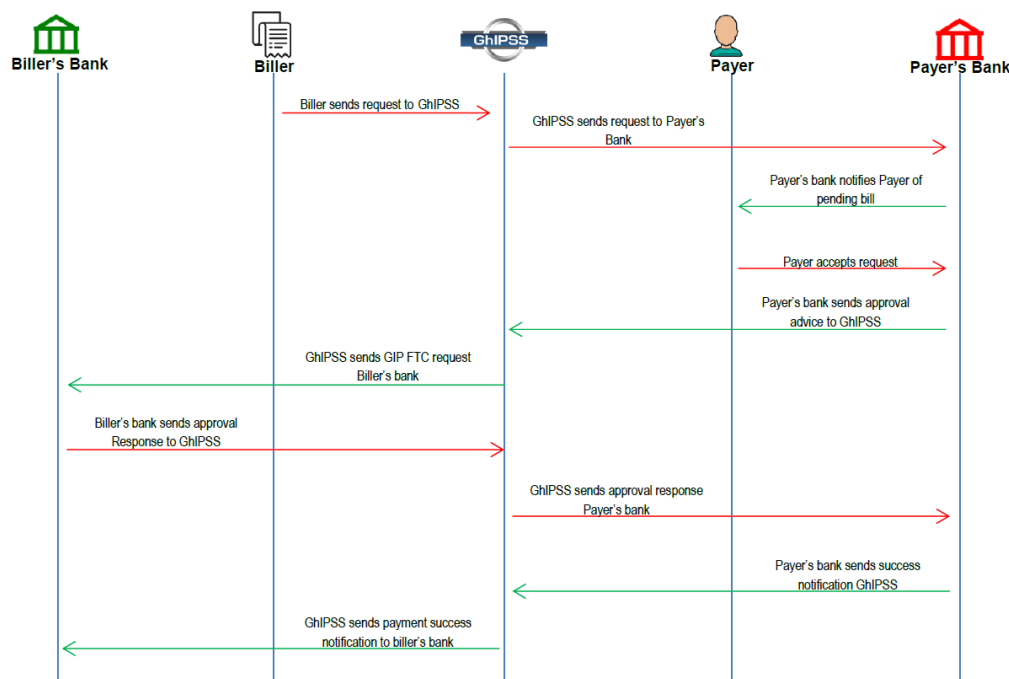
This API is exposed as REST services, TLS ciphered and secured through digital signatures.

1.1. REQUEST-TO-PAY

Request to Pay is a secure digital messaging and payment service. It is a simple and fast payment process that allows a beneficiary (Merchant, Biller etc) to initiate an electronic request for payment to a payer's (Customer's) account / wallet. The payer is notified immediately via their mobile device and can accept the request. If accepted, the requested amount is transferred instantly from the payer's account to the beneficiary's account / wallet and the beneficiary notified accordingly.

For each request, the payer has the choice to accept or decline to pay. Request-To-Pay is an overlay service on top of GhIPSS Instant Pay (GIP)

1.2. PROCESS FLOW



2. HTTP

- Base URL
 - API : <https://ghipss.com/ghipssapi/>
- Headers
 - Content-type: "application/json"
- Methods :
 - GET
 - POST
- HTTP Status Code :
 - 200 : Success
 - 420 : Application error
 - 500 : System error
- Data
 - All data is exchanged in JSON format.

3. JSON Fields

This section describes all the fields that the API uses in the different services it provides.

3.1. **institutionId**

- Type : Character String
- Length : 6 characters
- Description : Code given by GhIPSS to its member institutions (Banks, MMO ...)

3.2. **name**

- Type : Character String
- Length : Variable with a maximum of 128 characters
- Description: A name by which the individual/institution would be known.

3.3. **accountNumber**

- Type : Character String
- Length : Variable with a maximum of 32 characters
- Description: The account number which shall be used, later on, as the destination of payment on a GIP FTC.

3.4. **branchCode**

- Type : Character String
- Length : Variable with a maximum of 16 characters
- Description: A code identifying, if applicable, the branch of the institution to which the individual belongs. An "X" shall be used in case this information is not available/applicable.

3.5. **billerId**

- Type : Character String
- Length : Variable with a maximum of 64 characters
- Description: An ID chosen by a biller institution to be used as target of payments from the payer.

3.6. **payerInstitutionId**

- Type : Character String
- Length : 6 characters
- Description : Code given by GhIPSS to its member institutions (Banks, MMO ...).

3.7. payerAccountNumber

- Type : Character String
- Length : Variable with a maximum of 32 characters
- Description : The account number which shall be used, later on, as the source of payment on a GIP FTC.

3.8. billId

- Type : Character String
- Length : Variable with a maximum of 64 characters
- Description : ID of the bill that the Biller is requesting a payment for from the payer.

3.9. billDate

- Type : Character String
- Length : Variable with a maximum of 64 characters
- Description : Date of the bill for which the payment is being requested. The format of date in this field is ISO8601 Datetime.

3.10. billAmount

- Type : Character String
- Length : Variable with a maximum of 12 characters
- Description : Amount in Ghana Cedis of the bill for which a payment is being requested.

3.11. billDescription

- Type : Character String
- Length : Variable with a maximum of 128 characters
- Description : Extra details, about the bill for which a payment is being requested, that the billers wants to send to the payer to give more information/context/justification for the request that's being make.

3.12. requestSource

- Type : Character String
- Length : Variable with a maximum of 64 characters
- Description : A code indicating the source of the request of the enrolment. The possible values are the following :
 - IB : Internet Banking/Web Site of the institution
 - MB : Mobile application of the institution
 - USSD : USSD application of the institution
 - IVR : Interactive Voice Response



- BRANCH
- OTHER

3.13. requestTimestamp

- Type : Character String
- Length : Variable with a maximum of 32 characters
- Description : Date and time of the request expressed in ISO8601 Datetime format.

3.14. requestSignature

- Type : Character String
- Length : Variable with a maximum of 512 characters
- Description : A base 16 digital signature generated by the sender using the institution key along with HMAC technique with the algorithm SHA256. The data to be signed is the concatenation of values of all the fields sent in a message, except the signature, **In alphabetical order of the request's tag names**. In the API checks, another signature will be generated by the system and compared with the one received to confirm the integrity and authentication of the request.

3.15. callBackUrl

- Type : Character String
- Length : Variable with a maximum of 150 characters
- Description : The url of the notification service which will be called upon completing bill payment successfully in order to notify the biller that the bill payment successfully finished.

3.16. sessionId

- Type : Character String
- Length : Variable with a maximum of 12 characters
- Description: A session id that will be sent from payer's institution in case the payer accepted to pay the bill to identify the transaction, this session id will be used in the call for the FTC request.

3.17. billerTypeCode

- Type : Character String
- Length : Variable with a maximum of 2 characters
- Description: The ID must correspond to one of the biller types configured in the web portal. Ensure that the ID sent is included in the predefined list, the biller type will help in distinguishing between the different billers categories during fees calculation.

3.18. errorCode

- Type : Character String
- Length : Variable with a maximum of 16 characters
- Description : A code generated by the system to indicated the error that happened. The possible values are the following :
 - "200" : Operation was successful
 - "420.000" : Mandatory parameters were missing. Mainly the "state" parameter where applicable.
 - "420.001" : Mandatory JSON data fields were missing
 - "420.002" : The public key of the sending institution was not found in the system
 - "420.003" : The requested service is not authorized for the sending institution
 - "420.004" : The received signature is incorrect. It verification by the system failed.
 - "420.005" : The requested operation failed. Field "errorDescription" shall have more details on the failure.
 -
 - "420.007" : The biller id, for which an enrollment is being request, exists already in the system
 - "420.008" : The payer id, for which an enrollment is being request, exists already in the system
 - "420.009" : The bill, for which a payment is being request, exists already in the system
 - "420.010" : The destination id was not found. Field "errorDescription" shall have more details on the failure.

3.19. errorDescription

- Type : Character String
- Length : Variable with a maximum of 128 characters
- Description : A text in English giving a detailed description of the error that system detected

4. Request To Pay

This service is provided by GhIPSS in order to let billers request for payments.

4.1. Billers

4.1.1. Enrollment

- URL: <https://ghipss.com/ghipssapi/rtp/biller?state=add>
- Request Body:

```
{  
  
  "accountNumber": "0123456789012",  
  "billerId": "0618536308ss",  
  "billerTypeCode": "01",  
  "branchCode": "55002",  
  "callBackUrl": "https://callBackUrl.com",  
  "institutionId": "300311",  
  "name": "John DOE",  
  "requestSource": "IB",  
  "requestTimestamp": "2019-06-15T20:49:55.450+01:00",  
  "requestSignature": "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4a  
a7bc8ab2c8dd4"  
  
}
```

- Mandatory Field:
 - **institutionId**
 - **billerId**
 - **name**
 - **accountNumber**
 - **branchCode**
 - **requestSource**
 - **callBackUrl**
 - **requestTimestamp**
 - **billerTypeCode**
 - **requestSignature**

- Response Body

```
{  
  "errorCode": "200",  
  "errorDescription": "success"  
}
```

- Mandatory Field:
 - **errorCode**
 - **errorDescription**

4.1.2. Update

- URL: <https://ghipss.com/ghipssapi/rtp/biller?state=update>
- Request Body:

```
{  
  
  "accountNumber": "0123456789012",  
  "billerId": "0618536308ss",  
  "billerTypeCode": "01",  
  "branchCode": "55002",  
  "callBackUrl": "https://callBackUrl.com",  
  "institutionId": "300311",  
  "name": "John DOE",  
  "requestSource": "IB",  
  "requestTimestamp": "2019-06-15T20:49:55.450+01:00",  
  "requestSignature":  
    "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"  
  
}
```

- Mandatory Field:
 - **institutionId**
 - **billerId**
 - **name**
 - **accountNumber**
 - **branchCode**
 - **requestSource**
 - **callBackUrl**
 - **requestTimestamp**
 - **billerTypeCode**
 - **requestSignature**

- Response Body

```
{  
  "errorCode": "200",  
  "errorDescription": "success"  
}
```

- Mandatory Field:
 - **errorCode**



- **errorDescription**

4.1.3.Delete

- URL: <https://ghipss.com/ghipssapi/rtp/biller?state=delete>
- Request Body:

```
{  
  
  "billerId": "0618536308",  
  "institutionId": "300311",  
  "requestSource": "IB",  
  "requestTimestamp": "2019-06-17T23:37:41.445+01:00",  
  "requestSignature":  
    "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"  
  
}
```

- Mandatory Field:
 - **institutionId**
 - **billerId**
 - **requestSource**
 - **requestTimestamp**
 - **requestSignature**

- Response Body

```
{  
  "errorCode": "200",  
  "errorDescription": "success"  
}
```

- Mandatory Field:
 - **errorCode**
 - **errorDescription**

4.1.4.Query

- URL: <https://ghipss.com/ghipssapi/rtp/biller?state=query>
- Request Body:

```
{
```

```
"billerId": "6Y262Y26111",  
"institutionId": "300311",  
"requestSource": "IB",  
"requestTimestamp": "2019-06-15T15:26:07.194+01:00",  
"requestSignature":  
"745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"  
  
}
```

- Mandatory Field:
 - **institutionId**
 - **billerId**
 - **requestSource**
 - **requestTimestamp**
 - **requestSignature**

- Response Body

```
{  
  "institutionId": "300311",  
  "billerId": "6Y262Y26111",  
  "name": "KNUST",  
  "accountNumber": "0123456789012",  
  "branchCode": "55002",  
  "status": "N",  
  "callbackUrl": "https://callbackUrl.com",  
  "errorCode": "200",  
  "errorDescription": "success"  
}
```

- Mandatory Field:
 - **institutionId**
 - **billerId**
 - **name**
 - **accountNumber**
 - **branchCode**
 - **status** (N stands for Active biller, Y stands for Inactive biller)
 - **callbackUrl**
 - **errorCode**
 - **errorDescription**

4.1.5. Query Transaction Status

- URL: <https://ghipss.com/ghipssapi/rtp/queryStatus>
- Request Body:

```
{  
  
  "billId": "477828828545",  
  "billerId": "6Y262Y26111",  
  "payerAccountNumber": "0123456789012",  
  "requestTimestamp": "2019-06-15T15:26:07.194+01:00",  
  "requestSignature":  
    "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"  
}
```

- Mandatory Field:
 - **billId**
 - **billerId**
 - **payerAccountNumber**
 - **requestTimestamp**
 - **requestSignature**

- Response Body

```
{  
  "status": "P",  
  "reasonCode": "990",  
  "sessionID": "477828828545",  
  "errorCode": "200",  
  "errorDescription": "success"  
}
```

- Transaction Status possible values:
 - P: Request to pay is pending
 - A: Request to pay was accepted
 - D: Request to pay was declined
 - F: Request to pay was failed
 - U: Failed to reach payer's institution API to push the request
- Transaction Status reason codes possible values:
 - 000: In case the RTP was accepted
 - 100: In case the RTP was declined
 - 990: In case the RTP is pending on the payer's end



***: The reason value sent during the payer's notification and it should be predefined in the system.

***: The reason codes for transactions with status F (failed) should be consistent with the GIP action codes

4.1.6.List

- URL: <https://ghipss.com/ghipssapi/rtp/biller?state=list>
- Request Body:

```
{
  "institutionId": "300311",
  "requestSource": "IB",
  "requestTimestamp": "2019-06-15T15:26:07.194+01:00",
  "requestSignature":
  "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"
}
```

- Mandatory Field:
 - **institutionId**
 - **requestSource**
 - **requestTimestamp**
 - **requestSignature**

- Response Body

```
{
  "billers":[
    {
      "institutionId":"300311",
      "billerId":"0618536308",
      "status": "N",
      "name":"John DOE",
      "accountNumber":"0123456789012",
      "branchCode":"55002",
      "callBackUrl":"https://callBackUrl1.com"
    },
    {
      "institutionId":"300311",
      "billerId":"0618666666",
      "status":"N",
      "name":"LANAGEM",

```



```

    "accountNumber": "0123456789012",
    "branchCode": "55002",
    "callbackUrl": "https://callbackUrl2.com"
  },
  {
    "institutionId": "300311",
    "billerId": "6Y262Y26111",
    "status": "N",
    "name": "KNUST",
    "accountNumber": "0123456789012",
    "branchCode": "55002",
    "callbackUrl": "https://callbackUrl3.com"
  },
  {
    "institutionId": "300311",
    "billerId": "8888888888888888",
    "status": "N",
    "name": "SHOPRITE",
    "accountNumber": "0123456789012",
    "branchCode": "55002",
    "callbackUrl": "https://callbackUrl4.com"
  }
],
"errorCode": "200",
"errorDescription": "success"
}

```

- **Mandatory Field:**
 - **Billers**
 - **institutionId**
 - **billerId**
 - **status** (N stands for Active biller, Y stands for Inactive biller)
 - **name**
 - **accountNumber**
 - **branchCode**
 - **callbackUrl**
 - **errorCode**
 - **errorDescription**

4.2. Request

4.2.1. Request from Biller to GhIPSS

- URL: <https://ghipss.com/ghipssapi/rtp/request>
- Request Body:

```
{
  "billAmount": "123.45",
  "billDate": "2019-05-31",
  "billDescription": "Some bill that needs payment",
  "billId": "GH782882",
  "billerId": "6Y262Y26111",
  "payerAccountNumber": "0123456789012",
  "payerInstitutionId": "300311",
  "payerName": "John DOE",
  "requestTimestamp": "2019-06-17T17:20:24.726+01:00",
  "requestSignature":
    "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"
}
```

- Mandatory Field:
 - **billerId**
 - **billId**
 - **billDate**
 - **billAmount**
 - **payerInstitutionId**
 - **payerAccountNumber**
 - **payerName**
 - **billDescription**
 - **requestTimestamp**
 - **requestSignature**

- Response Body

```
{
  "errorCode": "200",
  "errorDescription": "success"
}
```



- Mandatory Field:
 - **errorCode**
 - **errorDescription**

4.2.2. Request from GhIPSS to Payer's Institution

- URL: [https:// \[URL\]](https://[URL]) Given by the Institution]
- Request Body:

```
{
  "billAmount": 123.45,
  "billDate": "2019-05-31",
  "billDescription": "Some bill that needs payment",
  "billId": "GH782882",
  "billerAccountNumber": "0123456789012",
  "billerId": "6Y262Y26111",
  "billerInstitutionId": "300311",
  "billerName": "KNUST",
  "payerAccountNumber": "0123456789012",
  "payerInstitutionId": "300311",
  "payerName": "John DOE",
  "requestTimestamp": "2019-06-17T17:20:24.726+01:00",
  "requestSignature":
    "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"
}
```

- Mandatory Field:
 - **billerId**
 - **billId**
 - **billDate**
 - **billAmount**
 - **billDescription**
 - **billerName**
 - **billerInstitutionId**
 - **billerAccountNumber**
 - **payerName**
 - **payerInstitutionId**
 - **payerAccountNumber**
 - **requestTimestamp**
 - **requestSignature**

- Response Body

```
{
  "errorCode":"200",
  "errorDescription":"success"
}
```

- Mandatory Field:
 - **errorCode**
 - **errorDescription**

- Check on Payer's Institution's Response:

1. **HTTP RESPONSE STATUS IS 200 :**

In this case we check the payload above to verify if **errorCode** field equals '200', once this check is done the process continues, the request is saved and marked then awaits for the payer's institution confirmation (accept/decline).

2. **HTTP RESPONSE STATUS IS DIFFERENT THEN 200 :**

If the payer's institution response was with an http response status different from 200(timeout, communication error...), the transaction is failed at this point, and the request is saved and marked as unreachable and can be viewed at the admin portal on **RTP UNREACHABLE PAYER BANKS report**.

4.2.3. Request from GhIPSS to Biller's Institution

- URL: [https:// \[URL Given by the GHIPSS\]](https:// [URL Given by the GHIPSS])
- Request Body:

```
{
  "AccountToCredit": "0123456789012",
  "AccountToDebit": "0123456789012",
  "Amount": 123.45,
  "ChannelCode": "915",
  "DestBank": 300311,
  "FunctionCode": "240",
  "NameToCredit": "KNUST",
  "NameToDebit": "John DOE",
  "Narration": "Some bill that needs payment",
  "OrigineBank": 300311,
  "SessionID": "211100743137",
  "TrackingNum": "211100",
  "datetime": "2019-05-31"
}
```

```
}
```

- Mandatory Field:
 - **NameToDebit**
 - **AccountToDebit**
 - **OrigineBank**
 - **NameToCredit**
 - **AccountToCredit**
 - **DestBank**
 - **Amount**
 - **datetime**
 - **TrackingNum**
 - **FunctionCode**
 - **SessionID**
 - **ChannelCode**
 - **Narration**

4.2.4. Notification from GhIPSS to Biller

- URL: [https:// \[URL\]](https:// [URL]) Given sent in the request]
- Response Body:

```
{
  "billAmount": 123.45,
  "billDate": "2019-05-31",
  "billDescription": "Some bill that needs payment",
  "billId": "GH782882",
  "billerId": "6Y262Y26111",
  "payerAccountNumber": "0123456789012",
  "payerInstitutionId": "300311",
  "payerName": "John DOE",
  "requestTimestamp": "2019-06-17T17:20:24.726+01:00",
  "sessionId": "050721214865",
  "status": "OK",
  "requestSignature":
    "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"
}
```



Once the payer institution responds to payment request with an approval (accept to pay the bill) or declines the payment, a notification will be sent to the biller using the callback url provided during its enrollment.

The “**status**” in biller notification can take the value “OK” in case the payer accepted to pay the bill, or “NOK” in case the payer refused to pay the bill.

NB: Once the payer institution responds to the payment request, the above notification will be sent to the biller (there is also a retrying mechanism until reaching number of retries specified in the admin).

The notification of the biller is considered successful when receiving a response code in the header

4.2.5. Notification from Payer’s institution to GHIPSS

- URL: <https://ghipss.com/ghipssapi/rtp/notification>
- Request Body:

```
{
  "billId": "7445672",
  "billerId": "JS785S4",
  "notificationTimestamp": "2021-06-15T18:06:56.312+01:00",
  "payerAccountNumber": "0123456789012",
  "payerBankId": "300311",
  "payerName": "Jhon Doe",
  "reasonCode": "01",
  "requestTimestamp": "2021-05-15T18:06:56.312+01:00",
  "sessionId": "050721214865",
  "status": "A",
  "requestSignature":
  "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"
}
```

- Mandatory Field:
 - **billerId**
 - **billId**
 - **payerAccountNumber**
 - **payerName**
 - **payerBankId**

- status (“A” to accept the bill payment, “D” to refuse the payment, F to indicate Failed RTP)
 - sessionId (will be used in the FTC request), this field is mandatory in case the status = ‘A’ otherwise it’s optional.
 - reasonCode:
 - If status is A(Accepted) the reason code must be 000
 - If status is D(Declined) the reason code must be 100
 - If status is F(Failed) the reason code value must be consistent with the GIP / MMI action code
 - requestTimestamp (request of the biller)
 - notificationTimestamp
 - requestSignature
- Response Body

```
{
  "errorCode": "200",
  "errorDescription": "success"
}
```

4.2.6.Name Enquiry

- URL: <https://ghipss.com/ghipssapi/rtp/nec/request>
- Request Body:

```
{
  "billerId": "JS785S4",
  "payerAccountNumber": "0123456789012",
  "payerInstitutionId": "300311",
  "requestSignature":
  "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60bbc4aa7bc8ab2c8dd4"
}
```

- Mandatory Field:
 - billerId
 - payerAccountNumber
 - payerInstitutionId
 - requestSignature

- Response Body

```
{
  "errorCode": "200",
  "errorDescription": "success",
  "payerName": "Test Payer"
}
```

4.2.7. Update Biller Status

- URL: <https://ghipss.com/ghipssapi/rtp/biller?state=status>
- Request Body:

```
{
  "billerId": "JS785S4",
  "institutionId": "JS785S4",
  "status": "N",
  "requestTimestamp": "2021-05-15T18:06:56.312+01:00",
  "requestSignature": "745f9ed2712a69d89998e03w170b3566257b8e18b82ef60b
bc4aa7bc8ab2c8dd4"
}
```

- Mandatory Field:
 - **billerId**
 - **institutionId**
 - **status** (“N” to mark the biller as normal or “Y” to mark the biller as cancelled = inactive biller)
 - **requestTimestamp**
 - **requestSignature**
- Response Body

```
{
  "errorCode": "200",
  "errorDescription": "success"
}
```